

A Study on The Relationship Between Reaction Time, Reactive Stress Tolerance, and Attention in Ncoe Athletes

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ABSTRACT

The main aim of this study was to find out if there existed a relationship among reaction time, reactive stress tolerance, and attention. A sample of 68 athletes belonging to National Centre of Excellence (NCOE) was taken from the Vienna Test System (VTS) database. The tests administered to the athletes were Simple reaction (RT/S1) for reaction time, Determination test (DT/S1) for reactive stress tolerance, and Cognitrone (COG/S4) for attention. Analysis of the data showed that reaction time and reactive stress tolerance, and reaction time and attention had a negative correlation with each other (-.529, -.320 respectively), while reactive stress tolerance and attention shared a positive correlation (.499). The correlations were found to be statistically significant. These findings help applied sport psychology practitioners and coaches to understand how different abilities of an athlete can affect performance.

KEYWORDS: reaction time, reactive stress tolerance, attention, Vienna test system.

INTRODUCTION

Vienna Test System (VTS)

VTS is a computerized test system for assessing psychological assessments such as personality test, reaction test, coordination test, perception test, and so on. VTS was developed by Schuhfried Company, founded by Dr. Felix Schuhfried in 1947. Under the VTS, a test called Determination Test is available for

assessing reactive stress tolerance and ability to react under complex stimulus conditions.

Reaction Test (RT)

The VTS contains various computerised collection of tests and assessments that can be used as a combination of tests or test alone. The system has 74 sets of

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different tests for work, health, and education which are categorised as Intelligence, General Ability, Special Ability, Personality Structure, Attitude and Interests and Clinical Tests. Reaction Test Analysis (RT) is one of the 8 General Ability Tests in the VTS. RT is a test of attention, measuring the reaction time to visual and auditory stimuli. The test is made for adults and children from six years of age. The VTS tests are mostly multi-lingual with majority in German and English. RT test requires the participants to respond to relevant stimuli. The response input is through a work panel. The participants have their finger on a gold button for the whole test time, allowing the participants to move to a black button when perceived a specific stimulus. Each test starts with an error sensitive practice phase. There are 10 forms of reaction tests (S1-S10) which differ in terms of modality, complexity, number of critical stimuli and test length. Tests from S1-S5 measures the reaction time (split into reaction time and motor time) to respond to simple and complex visual and auditory signals. For test form S1, the participants react when they see a yellow light, whilst for S2, they react to a 2000Hz tone.

Determination Test

The Determination Test or DT is one of the assessment tests in the VTS. It is considered to be a complex multiple stimulus reaction test which will

determine the reactive stress tolerance, attention deficits and reaction speed in the presence of rapidly changing and continuous visual and acoustic stimuli. The participants have to face a variety of colour and acoustic stimuli to which they have to respond by pressing coloured buttons (on the work panel of VTS) or by pressing the foot pedals according to a set of pre-determined rules. Therefore, the test of DT requires distinction between different colours and sounds, memorization of the stimulus ordering, response buttons on the work panel, and the correct decision of the specific responses. There are various forms of the test (16 forms) which is varied according to whether the test requires the use of foot pedals and the mode of presentation used. For the presentation mode, there are three ways: Reaction Mode (the presentation time is limited and the next stimulus is presented even when there is no response to the previous stimulus); Action Mode (the presentation time is unlimited and the next stimulus is presented only when there has been given a response to the previous stimulus); Adaptive Mode (the presentation speed depends on the speed of the response by the participants).

Cognitrone (COG)

Cognitrone (COG) is one of 8 General Ability Tests within the fully computerised Vienna Test System. COG is a general performance test where attention and concentration are assessed.

The COG is based on Reulecke's theory of concentration (1991) where it says that concentration is a state that can in principle be defined by three variables: energy (concentration uses energy); function (concentration is seen by performing a task); precision (the quality of task performance). In all the test forms of COG, the variable "energy" as described by Reulecke (1991) is measured by the time taken at a pre-set level of precision and function. COG is designed for test of adults and children of four years of age. The tests consist of abstract figures on which the participants make a judgment concerning their congruence. Items on the tests are presented on screen and the participants respond using either a computer board or the VTS work panel. Each test of COG begins with an animated instruction and an error sensitive practice phase. There are ten forms of COG (S1-S9 and S11) which are all different in terms of item difficulty, number of items, working time limit and number of figures in the display field.

Importance of stress tolerance, attention, and reaction time in sports

Stressors in sports differ from training to competitions and not able to handle these stressors can be psychologically challenging for many athletes which therefore can be an influencing factor in their sport performance. Therefore, effective stress management is essential

for the athletes to regulate competition related demands to facilitate optimal performance as well as to enhance psychological well-being. The ability to focus and pay attention to be able to concentrate on the performance of an athlete is one of the most important areas in the field of sports psychology. Therefore, it is important for an athlete to continue concentrating on the game instead of focusing on distractions like booze and buzz from the crowd. Concentrating and paying attention to the main objective of the game even when the athlete is losing would be ensured from a good attentional skill. Athletic performance can be affected by reaction time on which it can help the athlete react to each situation better and reduce risk of injuries. Furthermore, there is a close relationship between attention and reaction time. That is to say, the higher levels of attention result in a shorter reaction time, and the opposite is also true (Karwowski, 2006).

Objectives

The objective of this study is to assess the relationship between reaction time, reactive stress tolerance, and attention using Vienna Test System (VTS).

METHODOLOGY

Sample

The study aims to investigate the correlation among the various tests in the Vienna Test System (reaction time,

reactive stress tolerance, and attention) among Gymnasts and Hockey players. The sample was taken from 68 athletes (Hockey and Gymnastics) belonging to National Centre of Excellence (NCOE). The 68 athletes consisted of 38 males and 30 females.

Data collection

In the VTS available at the sports psychology department, the athletes performed on the specific tests provided. Three different tests were administered to the athletes - Reaction Test, Determination Test, and Cognitrone; they were allowed to give the tests on the VTS work panel. The environment was provided in a manner that the athletes were in a comfortable position for an effective result.

Tests Used

The following tests were administered using VTS:

1. Reaction Test: RT/S1 which measures the reaction time (split into reaction time and motor time); the athlete has to respond to a yellow light when seen at the screen.
2. Determination Test: DT/S1 which measures the reactive stress tolerance; the athlete has to respond to a reaction mode, where the stimulus is presented as distinction of different colours responses (whether or not there has been a response, the next stimulus will come).

3. Cognitrone: COG/S4 which measures the concentration or attention; the athlete has to complete the test in a limited time period.

RESULTS

Table 1: Descriptive Statistics and Correlations for Study Variables

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	1	2	3
1. Reaction time	68	284.	51.705	-		
2. Reactive stress tolerance	68	236.09	86.639	-.529*	-	
3. Attention	68	63.32	6.151	-.320*	.499*	-

* Correlation is significant at the 0.01 level (2-tailed).

The statistical analysis of the data was done using SPSS. Pearson's correlation test was applied on all the three variables. The score of Pearson's correlation between reaction time and reactive stress tolerance is -.529, which is a negative and a weak correlation, albeit statistically significant at the 0.01 level. Similarly, the Pearson's correlation between reaction time and attention is -.320, which is also a negative and weak correlation, but statistically significant at the 0.01 level. Likewise, the Pearson's correlation between reactive stress tolerance and attention is .499, which is a positive and weak correlation, but is statistically significant at the 0.01 level.

DISCUSSION

The aim of this study was to investigate the relationship between attention, reaction time, and reactive stress

tolerance among NCOE hockey players and gymnasts. The result of the study showed that there is a positive correlation between attention and reactive stress tolerance, and a negative correlation between reaction time and reactive stress tolerance, and reaction time and attention. The correlation values are weak, but they are statistically significant ($p < 0.01$). The result of this study and previous research conducted by Karwowski (2006) confirms that there is a negative correlation between attention and reaction time. Furthermore, results of the stress tolerance (DT) test conducted by Csaki et al. (2019) show that football players in the training group gave a relatively large number of incorrect responses suggesting that they might have been confused under pressure. From this it can be inferred that reactive stress

tolerance has an effect on attention. This study proved that there is a relationship among reaction time, reactive stress tolerance, and attention.

CONCLUSION

From the results obtained it can be concluded that there is a relationship between reaction time, reactive stress tolerance, and attention. It can be interpreted from the results that a negative, albeit weak relationship between reaction time and reactive stress tolerance, and reaction time and attention. Furthermore, it can be interpreted that there is a positive, albeit weak correlation between attention and reactive stress tolerance. Even though the correlations are weak, they are statistically significant ($p < 0.01$).

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